Configure IPv6 Addressing

1.Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

ANS:

Set a Static IPv6 Address (Windows)

Aim: To set a fixed IPv6 address on a computer and check it.

Steps

Step 1: Open Command Prompt

- Press Windows + R.
- Type cmd and press Enter.

Step 2: Check Wi-Fi Name

Type this command:

netsh interface ipv6 show interfaces

Step 3: Set IPv6 Address

Type:

netsh interface ipv6 add address "Wi-Fi" 2001:db8:1::10

Step 4: Check the Address

Type:

ipconfig

You can see the IPv6 address in your Wi-Fi details.

Real-Life Use

- A college can give a fixed IPv6 address to its computers.
- Office computers can use fixed addresses.
- Servers use fixed IP addresses so other devices can find them easily.

2.Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address.

Hint: Use the ping6 or ping -6 command for IPv6 testing.

ANS:

Configure a Unique Local IPv6 Address

Aim: To set a private IPv6 address on a computer and check the connection with another computer.

Steps (Windows)

Step 1: Open Command Prompt

- Press Windows + R.
- Type cmd and press Enter.

Step 2: Set IPv6 Address

Type this command:

netsh interface ipv6 add address "Wi-Fi" fd12:3456:789a::10

Step 3: Check IPv6 Address

Type:

ipconfig

You can see your IPv6 address starting with fd.

Step 4: Ping Another Device

Suppose another computer has this IPv6 address:

fd12:3456:789a::20

Type:

ping -6 fd12:3456:789a::20

Step 5: Check Result

- If you see Reply from, the connection is working.
- If you see Request timed out, check the network and firewall settings.

Real-Life Use

- Offices use private IPv6 addresses to connect computers.
- Colleges use them to connect computers in different labs.
- Private networks use them to share data safely.

3.Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect.

Constraint: Document the steps and the observed address behavior in a text file.

ANS:

Set Up a Temporary IPv6 Address

Aim: To check whether a temporary IPv6 address changes after restarting the computer or reconnecting Wi-Fi.

Steps (Windows)

Step 1: Open Command Prompt

- Press Windows + R.
- Type cmd and press Enter.

Step 2: Check IPv6 Address
Type:
ipconfig
Note down your IPv6 address.

Step 3: Enable Temporary IPv6 Address
Open Command Prompt as Administrator and type:
netsh interface ipv6 set privacy state=enabled

Step 4: Check the Address Again
Type:
ipconfig /all
Look for the Temporary IPv6 Address under your Wi-Fi adapter.

Step 5: Disconnect and Reconnect Wi-Fi
- Turn off Wi-Fi.
- Wait a few seconds.
- Turn Wi-Fi on again.
- Run ipconfig /all again.

Step 6: Restart Your Computer
After restarting, check the IPv6 address again.

Step 7: Write Your Observation

Situation	IPv6 Address
Before reconnecting	Note your address
After Wi-Fi reconnect	Check address again
After restart	Check address again

Real-Life Use
- Temporary IPv6 addresses help protect user privacy.
- They make it harder for websites to track the same device using one fixed address.
- They are commonly used by computers and mobile devices.

4.Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

ANS:

Restaurant Search Using IPv6 Address
Aim: To create a small program like Zomato that searches for restaurants and displays their IPv6 addresses.

Python Code
Restaurant names and IPv6 addressesrestaurants = { "Pizza Hut": "fd12:3456::1", "Dominos": "fd12:3456::2", "McDonalds": "fd12:3456::3", "KFC": "fd12:3456::4",

"Subway": "fd12:3456::5"}# Take restaurant name from usersearch = input("Enter restaurant name: ")# Search restaurantfound = Falsefor name, address in restaurants.items(): if search.lower() in name.lower(): print("Restaurant:", name) print("IPv6 Address:", address) found = Trueif not found: print("Restaurant not found!")

Steps to Run

1. Open VS Code.
2. Create a file named restaurant.py.
3. Copy and paste the code.
4. Run the Python program.
5. Enter a restaurant name, such as Pizza.
6. The program displays the matching restaurant and its IPv6 address.

Output

Enter restaurant name: Pizza

Restaurant: Pizza Hut
IPv6 Address: fd12:3456::1

Real-Life Use

- Zomato helps users search for restaurants.
- Users can enter a restaurant name.
- The program finds the restaurant and displays its IPv6 address.

5.Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

ANS:

IPv6 Subnetting Scenario Using ChatGPT

Aim: To create 3 IPv6 subnets and assign one IPv6 address to a computer.

Step 1: AI-Generated Scenario

A college has one main IPv6 network:

2001:db8:abcd::/48

The college needs 3 separate networks:

No.	Department	IPv6 Subnet
1	Computer Lab	2001:db8:abcd:1::/64

2	Library	2001:db8:abcd:2::/64
3	Admin Office	2001:db8:abcd:3::/64

Step 2: Assign IPv6 Address to Your System

Open Command Prompt as Administrator.

Check your Wi-Fi adapter name:

netsh interface ipv6 show interfaces

Assign an IPv6 address to Wi-Fi:

netsh interface ipv6 add address "Wi-Fi" 2001:db8:abcd:1::10

Step 3: Check Configuration

Type:

ipconfig

Check whether the IPv6 address appears under your Wi-Fi adapter.

Step 4: Real-Life Use

- Computer Lab uses its own IPv6 subnet.
- Library uses a separate IPv6 subnet.
- Admin Office uses another IPv6 subnet.
- This helps manage the college network easily.